

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Owned the search back end of a debate research SaaS platform that automates research and case-writing, building the *Python (FastAPI)* service on *AWS* alongside an *Agile* team of **10 engineers** for its **10K+ site visitors**
- Indexed **600K+ documents** in *Elasticsearch* behind *FastAPI* endpoints that return queries in **under 1 second**
- Designed a back-end text-retrieval algorithm in *Python* that validates source URLs and filters **IQR** outliers
- Created a de-duplication service that flagged **30%** of documents as identical, keeping the index lean as data grew
- Implemented recency-weighted ranking in the search API so evidence on the current debate topic returns first

Product Development Intern | Jetson

January 2025 – February 2025

- Ran **7 UI/UX testing iterations** on 3 features for **100K+ users**, asking action-oriented questions each round
- Cut user onboarding from **12 pages to 5** with a *Figma* redesign validated through **A/B testing** on 30 users
- Presented a **5-page report** to the CEO on strategies to optimize user engagement, communicating across teams

Data Analytics Intern | STEM E

June 2024 – July 2024

- Processed **200+ TikTok posts** from 3 years in *Python* with *pandas*, then charted trends in *Power BI* and *Excel*
- Spotted the key pattern: “science fun facts” posts earned 69.3% of all views while being just 7% of content
- Turned the insight into a business recommendation leadership adopted, raising engagement by **24% (8K+ views)**

PROJECTS

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Architected a *Python (FastAPI)* back-end service that streams live energy, water and waste readings from all 3 edge AI tools into **4 Random Forest models** and flags anomalies for the downstream dashboard in real time
- Exposed results through APIs to a *Next.js* dashboard where a **Gemma Voice LLM** delivers actionable steps
- Led 5 teammates to scope, drive and present the project end to end. **USAII Global Hackathon Finalist**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Wrote the *Python* back end of a scheduling product: a **5-step** algorithm that scans calendar availability, ranks open time slots and sends email confirmations to book meetings for groups of **30+ people** on Google Calendar
- Tested the back end with **14 unit and integration tests** to keep code quality high as new features shipped
- Built authentication for a lobby that verifies invited participants and queues follow-up meetings on *Firebase*
- Earned **8 startup licenses** and 2 signed LOIs (**\$3.5K**). **Winner, 2025 Congressional App Challenge**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built a back-end ingestion pipeline integrating 2 external APIs (*OpenAlex*, *Semantic Scholar*) that resolves each paper and backfills missing metadata into the *PostgreSQL* database for fast, reliable downstream *SQL* queries
- Modeled paper records in *PostgreSQL (Supabase)*, writing *SQL* queries that filter by field, year and citations
- Deployed a *Claude Sonnet 4.5 RAG* agent on the back end that retrieves session-scoped vector documents

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Built a *PyTorch* training pipeline in *Python* for an *EfficientNetV2-S* model on **15K+ images (96.13% accuracy)**
- Wrote a synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) that generated **9K** training images
- Documented the model’s methodology on *arXiv* with 2 *Kaggle* datasets. **2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks/Platforms: FastAPI, Flask, PostgreSQL, MySQL, Elasticsearch, Supabase, Firebase, AWS, Node.js

Concepts/Practices: API Development, REST APIs, Data Pipelines, Data Ingestion, Unit Testing, Authentication

DevTools: Docker, CI/CD, Git, GitHub, VS Code, Claude Code, Codex, Figma, Bun, Arduino IDE, Kaggle